Robinhood Marketing Channel Performance: Research and Conclusion

A full summary of our analysis in Python, SQL, Excel and Power BI. Jan to Jun 2025 results. Amounts are in rupees (₹96 = $1, an example rate).

1. The Problem

Robinhood lets people open a free account. Some of them later pay for a paid plan (Robinhood Gold). From January to June 2025, the company spent money on ads in four places: search ads, social media ads, banner ads (display) and finance blogs.

The decision: the head of marketing (CMO) must split the budget for the next six months. Which ads should get more money, which less, and which none?

Our job: find out which ads bring paying customers at the best cost, compare them with free signups, and recommend a budget split.

2. The Data

- **ad_spend.csv:** daily spend, views and clicks for each ad (1,046 rows, 1,039 after removing 7 repeated rows).
- **signups.csv:** one row per new user (18,840 rows), with the channel, the ad, the payment date, the plan and revenue in the first 90 days.

Data checks and fixes

- 7 repeated spend rows were removed, so spend was not counted twice.
- Channel names were written differently in the two files (Paid Search and paid_search). We made a matching key.
- 4,574 signups have no ad. They are organic and referral, so they have no ad spend.
- All 16,585 free users have ₹0 revenue, so revenue totals are safe.
- 9 signup rows have a channel and ad that do not match. They are 0.05% of the data and do not change the results.
- The SQL, Excel and Power BI spend total is a little under 1% lower than the Python total. The cause was not found. It does not change any ranking.

3.Analysis Build on :

1. **Python (pandas):** cleaned the data, joined spend and signups, and calculated click rate, cost per click, cost per signup, cost for each paying customer and ROAS. Compared paid and free channels, and looked at the six-month trend.
2. **SQL (SQL Server):** repeated the analysis with simple queries, plus extra questions on plans, days to pay and monthly trends.
3. **Excel:** prepared the data (removed repeats, matching names, rupee columns, month columns).
4. **Power BI:** built a 2-page dashboard. Page 1 shows the big picture. Page 2 shows ad by ad.

4. What We Found

The big picture

- Money spent on ads: about ₹1.20 crore.
- Money earned in the first 90 days: ₹89.7 lakh from all channels, and ₹66.7 lakh from ad signups.
- 18,840 signups: 2,255 paid (12%) and 16,585 did not pay.
- Most signups came in March (3,241). Most paying signups came in May (407).
- paid_search brought the most paying signups (897).

Ad by ad (first 90 days)

- **brand_search:** spent ₹9.8 lakh, earned ₹18.7 lakh, 465 paying customers. Cost for each paying customer ₹2,105. Money back per ₹1: ₹1.91.
- **retargeting:** spent ₹12.5 lakh, earned ₹16.7 lakh, 418 paying customers. ₹2,983 each. Money back per ₹1: ₹1.34.
- **generic_search:** spent ₹34.5 lakh, earned ₹16.3 lakh, 432 paying customers. ₹7,986 each. Money back per ₹1: ₹0.47.
- **finance_blogs:** spent ₹14.8 lakh, earned ₹6.1 lakh, 162 paying customers. ₹9,122 each. Money back per ₹1: ₹0.41.
- **prospecting_video:** spent ₹32.6 lakh, earned ₹8.4 lakh, 213 paying customers. ₹15,286 each. Money back per ₹1: ₹0.26.
- **display_network:** spent ₹15.8 lakh, earned ₹0.5 lakh, only 8 paying customers. ₹1,97,456 each. Money back per ₹1: ₹0.03.

Only brand_search and retargeting earned back more than they cost within 90 days.

The main surprise: cheap signups are not cheap paying customers

- prospecting_video is 2nd cheapest for a signup (about ₹507), but 5th for a paying customer (₹15,286).
- It brought 6,481 signups, but only 213 paid (about 3 out of 100). retargeting brought 2,466 signups and 418 paid (about 17 out of 100).
- generic_search moves up from 5th to 3rd when we look at paying customers.
- So ads must be judged by the cost of a paying customer, not the cost of a signup.

Do paid signups pay more than free ones? (Out of every 100 signups, how many pay)

- paid_search 21, referral 20, affiliate 19, organic 10, paid_social 7, display 3.
- Search ads and finance blogs bring users who pay as often as referral users. Social and banner ads bring users who pay less often than organic users.
- The social media number mixes two very different ads: retargeting (about 17) and prospecting_video (about 3).

Clicks

- brand_search has the best click rate (about 4.5 clicks per 100 views). prospecting_video has the cheapest click (about ₹48). display_network has the worst click rate (about 0.1 per 100 views) and the most expensive click (about ₹202).

Plans and speed

- Most customers choose the monthly plan. The yearly share is about 30% for brand_search and retargeting, 28% for prospecting_video, 25% for generic_search and 24% for finance_blogs.
- People take about 3 days to start paying, for every ad.

What changed over six months

- display_network stopped after May (about ₹3.4 to ₹3.7 lakh a month until April, about ₹1.6 lakh in May, nothing in June). The reason is not in the data. It was the worst ad, so it was probably paused for poor results.
- The other five ads spent about the same every month.
- The cost for each paying customer stayed steady: brand_search ₹1,900 to ₹2,300, retargeting ₹2,800 to ₹3,300, generic_search ₹7,300 to ₹9,500, finance_blogs ₹7,800 to ₹11,000. prospecting_video was ₹21,800 in January, then ₹14,200 to ₹15,100.

Do ads that spend more get more paying customers? No. generic_search and prospecting_video spend the most, but brand_search and retargeting bring about as many paying customers for far less money.

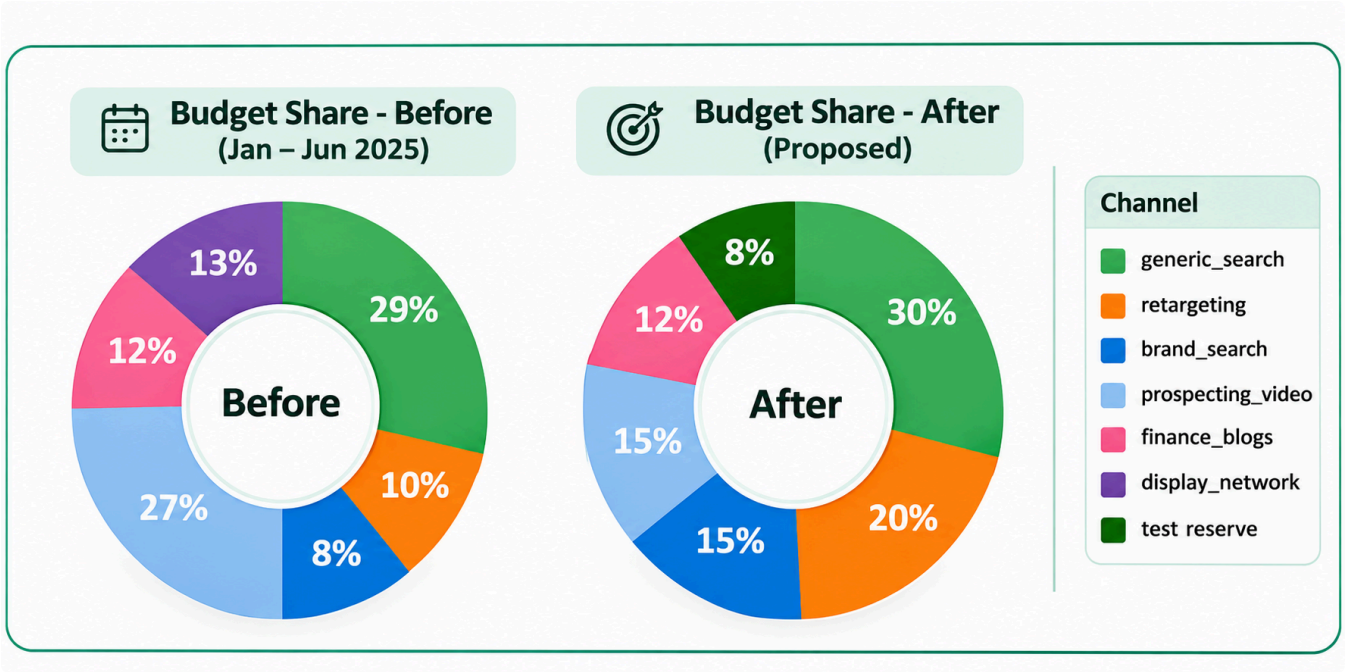
5. Conclusion

1. **brand_search and retargeting are the best ads.** They are the cheapest way to get a paying customer and the only two that earned back more than they cost.
2. **prospecting_video looks good only on signups.** Its signups rarely pay, so each paying customer costs about 7 times more than from brand_search.
3. **display_network is the worst ad.** It cost ₹1,97,456 for each paying customer and was already stopped after May.
4. **generic_search and finance_blogs lose money in 90 days, but they are not hopeless.** They bring paying customers at a medium cost, so keep them under watch.
5. **Judge ads by the cost of a paying customer** (and ROAS), not by the cost of a signup.
6. **Quality matters more than quantity.** More signups do not mean more money.

6. Recommendation: Budget Split for the Next Six Months

An example split to start from (the CMO can change it):

- **generic_search: 30%** (keep, watch the cost)
- **retargeting: 20%** (increase)
- **brand_search: 15%** (increase)
- **prospecting_video: 15%** (reduce, keep some to bring new people to retargeting)
- **finance_blogs: 12%** (keep, watch the cost)
- **test reserve: 8%** (to try new ideas)
- **display_network: 0%** (stop)



About a quarter of the budget moves away from prospecting_video and display_network. Most of it goes to retargeting and brand_search, and 8% goes to a test reserve.

Rough estimate of the result: if we cut prospecting_video by half, stop display_network, put the freed money into brand_search and retargeting, and lose 30% of results because new money works less well, paying signups could go from 2,255 to about 3,050 (+35%). This is only an estimate. Real results may be lower.

Assumptions: the exchange rate is ₹96 for $1. The target cost for each paying customer is ₹4,800. We do not know the total budget, so we assumed it stays the same as before.

7. Risks and Limits

- **Only 90 days of money earned is counted.** Some customers keep paying for longer, especially on yearly plans. So the slower ads may look worse than they really are.
- **Some brand_search signups may have happened anyway.** People searching for Robinhood by name may have signed up without the ad.
- **Results may drop when we spend more.** brand_search and retargeting reach a limited group of people.
- **Each signup is linked to only one ad.** A person may have seen several ads.
- **June signups are still new.** They have had little time to start paying.
- **The data shows what happened, not why.** For example, we do not know why display_network stopped.

8. Next Steps

1. **Test before spending everything.** For example, cut brand_search spend by 10% in one region for four weeks and see if paying customers drop.
2. **Watch 3 numbers each week:** cost for each paying customer, how many signups pay, and spend on each ad. Raise an alarm if retargeting goes above about ₹4,800 for a paying customer.
3. **Ask the CMO:** What is the total budget? What is the target cost for a paying customer? Is any ad a must-keep?
4. **Look at revenue beyond 90 days** to judge the yearly-plan customers fairly.